INSTITUT PENDIDIKAN INDONESIA

Program Studi Pendidikan Teknologi Informasi

RPS MATA KULIAH SISTEM OPERASI

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Referensi: MIT · Stanford · CMU · UC Berkeley · Georgia Tech

A. GAMBARAN UMUM MATA KULIAH

Mata kuliah ini mempelajari **bagaimana komputer mengelola sumber daya perangkat keras** (CPU, memori, penyimpanan, jaringan) serta **menyediakan layanan bagi program-program yang berjalan di atasnya**.

Atribut	Deskripsi
Nama Mata Kuliah	Sistem Operasi / *Operating Systems*
Kode Referensi	
Jenjang	S1
SKS Umum	3 SKS
Prasyarat	Arsitektur Komputer, Pemrograman C/C++, Struktur Data, Organisasi Komputer

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B. TUJUAN PEMBELAJARAN (*LEARNING OUTCOMES*)

Setelah menyelesaikan mata kuliah ini, mahasiswa mampu:

1. **Memahami prinsip desain** sistem operasi modern secara komprehensif
2. **Menganalisis dan mengevaluasi** mekanisme manajemen proses, memori, dan I/O
3. **Mengimplementasikan** komponen OS dasar seperti scheduler, page table, dan sistem file
4. **Mengidentifikasi trade-off** dalam desain OS: performa vs. keamanan, simplisitas vs. fleksibilitas
5. **Membaca dan mendiskusikan** paper penelitian sistem operasi tingkat lanjut (untuk tingkat graduate)

6. **Men-debug** masalah kompleks di level sistem (kernel panic, race condition, deadlock)

C. KURIKULUM & TOPIK UTAMA

1: PENGANTAR & FONDASI OS

Topik:

- Definisi, fungsi, dan struktur sistem operasi
- Sejarah perkembangan OS: Batch system → Multiprogramming → Time-sharing → Modern OS
- Arsitektur OS: Monolithic Kernel, Microkernel, Hybrid Kernel, Exokernel
- Mode operasi: Kernel Mode vs. User Mode
- System Calls: antarmuka antara aplikasi dan OS
- Bootstrap & booting process

Referensi:

- MIT 6.S081 — *Lecture 1: Introduction and Examples* (menggunakan OS xv6 berbasis RISC-V)
- Stanford CS140 — *Introduction to OS structure*

2: MANAJEMEN PROSES & *THREAD*

Topik:

- Konsep Proses: PCB (Process Control Block), state diagram (new, ready, running, blocked, terminated)
- Operasi proses: fork(), exec(), wait(), exit()
- Proses vs. Thread: perbedaan, keuntungan, dan kerugian
- Thread model: User-level threads vs. Kernel-level threads
- Thread library: POSIX Threads (pthreads)
- Context switching: mekanisme dan overhead

Subtopik Penting:

- Many-to-One, One-to-One, Many-to-Many threading models
- Green threads dan goroutines (modern implementation)
- Proses zombie dan orphan

3: PENJADWALAN CPU (*CPU SCHEDULING*)

Topik:

- Kriteria penjadwalan: CPU utilization, throughput, turnaround time, waiting time, response time
- Algoritma penjadwalan:
  - **FCFS** (First Come First Served)
  - **SJF** (Shortest Job First) — preemptive (SRTF) & non-preemptive
  - **Priority Scheduling**
  - **Round Robin (RR)**
  - **Multilevel Queue Scheduling**
  - **Multilevel Feedback Queue (MLFQ)**
  - **Completely Fair Scheduler (CFS)** — digunakan Linux
  - **Lottery Scheduling** dan **Stride Scheduling**
- Real-Time Scheduling: Rate Monotonic, Earliest Deadline First (EDF)
- Multiprocessor Scheduling: Symmetric vs. Asymmetric

Referensi: OSTEP (Three Easy Pieces) — *Chapter: CPU Scheduling & MLFQ*

4: SINKRONISASI & KONKURENSI

Ini adalah salah satu modul terberat dan terpenting dalam OS.

Topik:

- Race condition dan critical section problem
- Solusi software: Peterson's Algorithm
- Hardware primitives: TestAndSet, CompareAndSwap (CAS), FetchAndAdd
- Mekanisme sinkronisasi:
  - **Mutex Lock** (Mutual Exclusion)
  - **Semaphore** (binary & counting) — Dijkstra
  - **Monitor** dan **Condition Variables**
  - **Spinlock** vs. **Sleeping Lock**
  - **Read-Write Lock**
- **Deadlock**:
  - Empat kondisi Coffman: Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait
  - Deadlock Prevention, Avoidance (Banker's Algorithm), Detection & Recovery
- **Livelock** dan **Starvation**
- Lock-free dan wait-free data structures

Referensi: Stanford CS140 — *Concurrency*, MIT 6.S081 — *Labs: Locks & Multithreading*

5: MANAJEMEN MEMORI

Topik:

- Konsep dasar: address space, logical vs. physical address
- Memory allocation: contiguous, non-contiguous
- **Segmentasi**: segment table, segment fault
- **Paging**:
  - Page table, page size, internal fragmentation
  - Multi-level page tables (2-level, 3-level, x86-64 menggunakan 4-level)
  - TLB (Translation Lookaside Buffer) — hardware cache untuk page table
  - TLB miss handling: hardware-managed vs. software-managed
- **Virtual Memory**:
  - Demand paging dan page fault handling
  - Page replacement algorithms: FIFO, Optimal (OPT), LRU, Clock (Second-Chance), LRU Approximation
  - Thrashing dan working set model
  - Copy-on-Write (COW)
- **Memory-Mapped Files** (mmap)
- **Kernel Memory Allocation**: slab allocator, buddy system

6: SISTEM FILE (*FILE SYSTEMS*)

Topik:

- Konsep file dan direktori: path, file descriptor, inode
- Interface sistem file: open, read, write, close, lseek, stat
- **Implementasi sistem file**:
  - FAT (File Allocation Table): FAT12/16/32
  - Unix File System (UFS/FFS): inode-based, direct/indirect block pointers
  - **ext2/ext3/ext4** (Linux)
  - **NTFS** (Windows)
  - **APFS** (macOS)
  - **ZFS** — Copy-on-Write filesystem
  - **Log-Structured File System (LFS)**
  - **Journaling File Systems** — crash recovery
- Directory implementation: linear list vs. hash table vs. B+ tree
- Disk allocation methods: contiguous, linked, indexed
- Free space management: bitmap, free list
- **RAID** (Redundant Array of Independent Disks): RAID 0, 1, 5, 6, 10

- **Crash Consistency**: write-ahead logging, fsck

Referensi: OSTEP — *Persistence: File Systems*, MIT 6.S081 — *Lab: File System*

7: MANAJEMEN I/O & *STORAGE*

Topik:

- Struktur I/O: polling, interrupt-driven I/O, DMA (Direct Memory Access)
- Device driver architecture
- Disk structure: platters, tracks, sectors, cylinders
- Disk scheduling algorithms: FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK
- Storage technologies: HDD, SSD (NAND Flash), NVMe
- **Spooling** dan buffering
- Karakteristik I/O: block device vs. character device

8: KEAMANAN & PROTEKSI OS

Topik:

- Protection domain dan access matrix
- Access Control List (ACL) vs. Capability list
- **Security threats**: malware, rootkit, buffer overflow, privilege escalation
- Security mechanisms:
  - **ASLR** (Address Space Layout Randomization)
  - **Stack Canaries**
  - **NX/XD bit** (No-Execute)
  - **Sandboxing** dan **containerization**
  - **Secure Boot** dan trusted computing
- OS-level security: SELinux, AppArmor
- **Virtualization security** dan hypervisor isolation

9: VIRTUALISASI

Topik:

- Konsep virtual machine (VM)
- **Hypervisor**: Type 1 (Bare-metal: VMware ESXi, KVM, Xen) vs. Type 2 (Hosted: VirtualBox, VMware Workstation)

- Teknik virtualisasi CPU: full virtualization, paravirtualization, hardware-assisted (Intel VT-x, AMD-V)
- Memory virtualization: shadow page tables, EPT (Extended Page Tables)
- **Containers** vs. Virtual Machines: Docker, LXC, namespaces, cgroups
- OS-level virtualization: chroot, seccomp

10: TOPIK LANJUTAN

Diambil dari Stanford CS240 dan CMU 15-712:

- **Distributed Systems** dasar: RPC (Remote Procedure Call), consistency model
- **Exokernel** dan **Unikernel** design
- **Persistent Memory** (NVDIMM) dan dampaknya pada OS design
- **OS untuk Machine Learning**: sistem file untuk distributed training, GPU memory management
- **Microkernels**: L4, seL4 (formally verified microkernel)
- **Capability-based security** systems

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4. PRAKTIKUM & *LAB ASSIGNMENTS*

Pelaksanaan praktikum menggunakan sistem lab berbasis implementasi nyata. Sebagai acuan MIT 6.S081 memiliki **11 labs** sepanjang semester:

No	Lab Name	Topik
1	Utilities	Unix utilities dasar menggunakan system calls
2	System Calls	Tambahkan system call ke xv6
3	Page Tables	Manipulasi page table di kernel
4	Traps	Interrupt handling dan traps
5	Copy-on-Write Fork	Implementasi COW fork
6	Multithreading	User-level thread switching
7	Network Driver	Driver jaringan berbasis interrupt
8	Locks	Peningkatan paralelisme dengan fine-grained locking
9	File System	Implementasi large files dan symbolic links

No	Lab Name	Topik
10	Mmap	Memory-mapped file implementation
11	Final Project	Proyek OS mandiri

Platform yang digunakan:

- **xv6-RISC-V** (MIT) — sistem operasi edukasi berbasis RISC-V
- **Pintos** (Stanford/CMU) — OS sederhana berbasis x86 untuk edukasi
- **NachOS** (UC Berkeley) — Not Another Completely Heuristic Operating System

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5. REFERENSI & BUKU TEKS UTAMA

Buku	Penulis	Keterangan
Operating Systems: Three Easy Pieces (OSTEP)	Remzi H. & Andrea C. Arpaci-Dusseau (UW-Madison)	Free online, sangat direkomendasikan, digunakan di MIT & CMU
Modern Operating Systems (5th Ed.)	Andrew S. Tanenbaum & Herbert Bos (VU Amsterdam)	Standard textbook klasik, digunakan di ratusan universitas dunia
Operating System Concepts (10th Ed.) — "Dinosaur Book"	Silberschatz, Galvin, Gagne	Paling umum digunakan di Indonesia
xv6: A Simple Unix-like Teaching OS	Russ Cox, Frans Kaashoek, Robert Morris (MIT)	Free PDF, digunakan di MIT 6.S081
The Linux Programming Interface	Michael Kerrisk	Referensi implementasi Linux mendalam
Computer Systems: A Programmer's Perspective (CS:APP)	Bryant & O'Hallaron (CMU)	Digunakan di CMU 15-213 sebagai fondasi

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6. TOOLS & TEKNOLOGI YANG DIGUNAKAN

- **Bahasa Pemrograman:** C (primer), C++, x86/RISC-V Assembly
- **OS Platform:** Linux (Ubuntu/Debian), xv6

- **Emulator:** QEMU (untuk menjalankan OS kernel secara virtual)
- **Debugger:** GDB (GNU Debugger) dengan ekstensi kernel
- **Tools:** Make, objdump, strace, valgrind, perf
- **Version Control:** Git

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7. METODE PENILAIAN

Berdasarkan struktur penilaian dari MIT dan Stanford:

Komponen	Bobot
Lab Assignments / Programming Projects	40–50%
UTS	20–25%
UAS	25–30%
Reading/Discussion Participation	10–15%
Homework / Quizzes	5–10%

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8. KAITAN DENGAN MATA KULIAH LAIN

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Arsitektur Komputer ──────┐
Pemrograman C             ├──► SISTEM OPERASI ──►  Cloud Computing
Struktur Data             │                    ──►  Keamanan Jaringan
Matematika Diskrit ───────┘                    ──►  Distributed Systems
                                               ──►  Compiler Design
                                               ──►  Embedded Systems
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9. RESOURCE ONLINE GRATIS TERPERCAYA

- **MIT OpenCourseWare 6.S081** — https://pdos.csail.mit.edu/6.828/
- **OSTEP Book (Free PDF)** — https://ostep.org
- **xv6 Book (MIT)** — https://pdos.csail.mit.edu/6.828/xv6/book-riscv-rev2.pdf
- **UC Berkeley CS162 (YouTube)** — Lecture videos tersedia di YouTube
- **Stanford CS140 (Online)** — Materi tersedia di web.stanford.edu
- **OSDev Wiki** — https://wiki.osdev.org (referensi implementasi OS)

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10. RELEVANSI & PROSPEK KARIR

Pemahaman mendalam tentang Sistem Operasi membuka karir di bidang:

- **Systems Engineer / Kernel Developer** (Linux Foundation, Google, Meta)
- **Cloud Infrastructure Engineer** (AWS, Google Cloud, Azure)
- **Embedded Systems Developer** (automotive, IoT, robotics)
- **Security Researcher** (vulnerability analysis, exploit development)
- **Platform Engineer / SRE** (Site Reliability Engineering)
- **Compiler Engineer dan Runtime Developer**
- **Virtualization & Container Engineer** (Docker, Kubernetes internals)